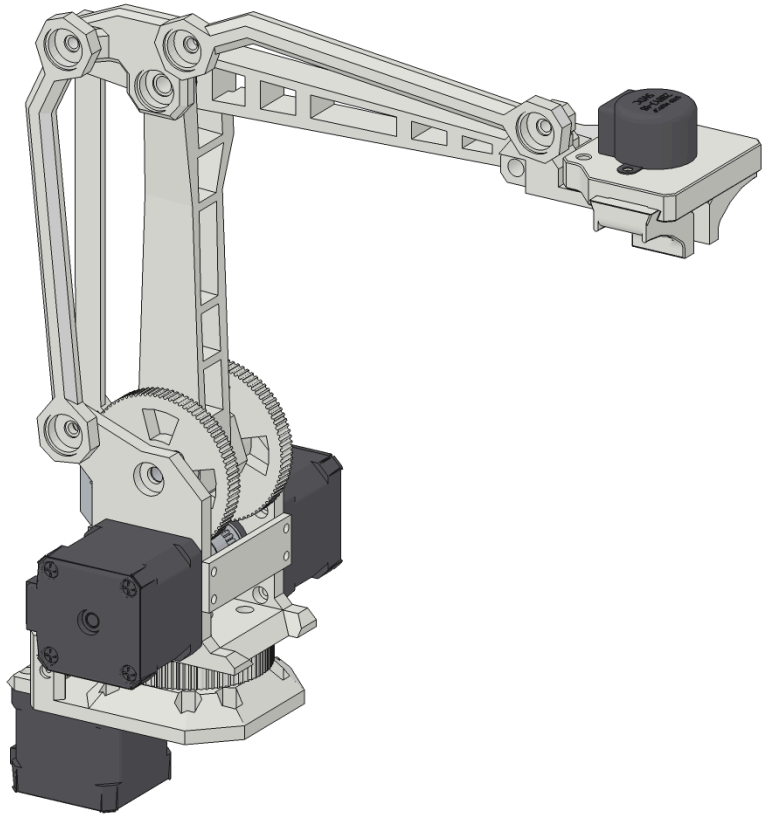


Mechy Demo bam-v2

[1] No catch simulation for this variant (Mechy no longer runs the catch simulation).

REACH

0.334 m computed

PURCHASED PARTS

$131 estimate

FINDINGS

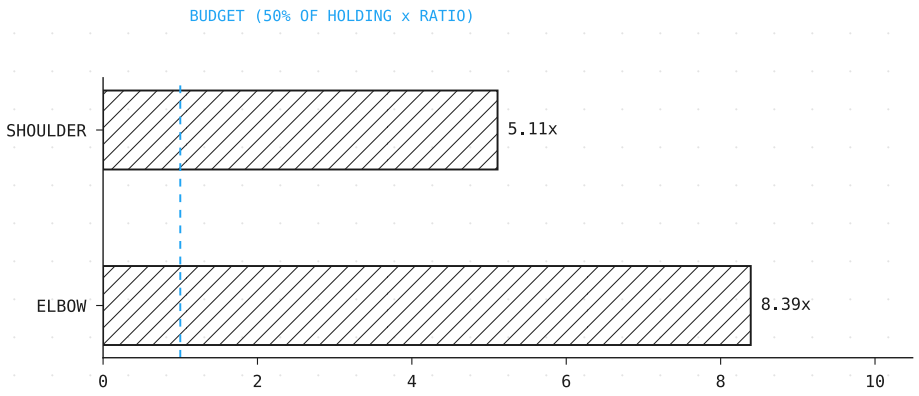
[1] Weakest motor: shoulder at 5.11x the 50% torque budget, arm stretched horizontally with the ball (computed).

2 Requirements vs results

REQUIREMENT	TARGET	RESULT	STATUS	SOURCE
Catch rate	>= 80%	not simulated	OPEN	no catch simulation for this variant
Ball fits the cup	ball 67 mm	cup 95 mm inside the liner	PASS	design/bam-v2/params.yaml
Torque budget, shoulder motor	>= 1.0x of 50% budget	5.11x (0.248 of 1.266 N m)	PASS	computed (ArmModel, mass_props.json)
Torque budget, elbow motor	>= 1.0x of 50% budget	8.39x (0.151 of 1.266 N m)	PASS	computed (ArmModel, mass_props.json)
Vision	? fps, 0 ms	modelled in the sim (camera noise + latency)	INFO	requirements.yaml

3 Torque margins

STANDARDS RULE 4: HOLDING X RATIO X 50%



MOTOR	RATIO	STATIC NM	BUDGET NM	MARGIN	MAX DEG/S
base	5.6	0.000	1.266	–	80
shoulder	5.6	0.248	1.266	5.11x	80
elbow	5.6	0.151	1.266	8.39x	80
wrist	4.5	0.000	1.012	–	100

NOTES

[1] Static case: arm stretched horizontally with the ball in the cup, the worst gravity load (computed from the CAD masses).

[2] With belt pass-through each motor holds the motor-side torque C^-T tau, not the joint torque (see mass_props.json motor_map).

[3] Budget = holding torque x ratio x 50% (standards rule 4). Holding torque is assumed until the motor model is confirmed.

MECHY

SHEET TITLE: TORQUE MARGINS

PROJECT	VARIANT	SCALE	DATE	SHEET
Mechy Demo	BAM-V2	NTS	2026-10-30	3 / 6

DATE	DECISION	WHY	EVIDENCE
2026-10-30	Correct requirements to 600 g and gate new CAD on capacity screening and user concept selection (why: explicit payload answer supersedes 100 g; existing performance is not verified at 600 g).		.mechy/digest.md
2026-10-30	Add an independent Uno built-in LED blink test using board id uno; default to 1 second on/off and a serial heartbeat (why: the user explicitly requests a board test, independent of the arm firmware handled by their friend).		inbox/20261030-102718.md
2026-10-30	Supersede queued demo tasks T101–T104 and blocked T108 with T111 inventory completion, T112 missing prior art and T113 concept selection (why: baseline T105 is absent, T106 lacks a completion summary, and 600 g screening fails; no heavy CAD before selection).		research/payload-600g/summary.md
2026-10-30	Schedule T110 to recheck the connected Uno and restore the existing blink only if needed (why: the new inbox request postdates the completed T109 test; reuse its verified defaults).		inbox/20261030-111114.md

5 Open decisions and gaps

OPEN DECISIONS

[1] the 4th motor/driver — vendor rail module or a wrist servo (spare). (design/bam-v2/summary.md)

[2] Choose a concept to develop for the 600 g arm. A is the conventional first option; A and B are the proposed CAD/simulation comparison pair, with C available if you prefer counterweights. No new CAD until your pick. All retain four owned motors, wrist roll, MG90S, black finish, X1E fit target and PETG exceptions. Grip, exact motor/step-rate capability and delivered cost below $100 remain unverified; choosing a concept does not approve fabrication or a budget increase. (options/arm-600g-concept.yaml)

NOT VERIFIED

[1] belt tensioning and tooth skipping, printed-surface quality, joint stiffness/backlash, infill-based masses (estimate — weigh the parts), home sensor positions (estimates, calibrate). Open decisions: the 4th motor/driver — vendor rail module or a wrist servo (spare). (design/bam-v2/summary.md)

6 Where the numbers come from

QUANTITY	VALUE	LABEL	SOURCE
Catch rate	–	COMPUTED	–
Motor holding torque	? N m	ASSUMED	hw/motors.yaml: –
Motor torque curve	0 points	ESTIMATE	hw/motors.yaml: –
Step rate cap	4000 Hz	ASSUMED	hw/drivers.yaml: default
Printed mass	206 g	COMPUTED	volume x density x shell/infill (estimate until weighed)
Purchased parts	$131	ESTIMATE	design/bam-v2/bom.csv
Cup liner restitution	0.15	ESTIMATE	params.yaml catcher.liner = net_pocket; drop-test it
Vision noise	4 mm	ASSUMED	requirements.yaml vision.noise_m

[1] Labels: COMPUTED = derived by the engine from other files; DATASHEET = vendor figure; ASSUMED / ESTIMATE = not measured, change it when you measure.